

A Common-Ambient Union-Shell Holdout for Transported Prime-Shell Labels

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Abstract

The preceding two-way budget decomposition for the twin-prime dynamical program still compared different physical shell operators and row-specific off-overlap completions. We place each adjacent shell pair in one common union ambient operator, fit aligned directional targets only on the overlap, and reserve the exclusive shell pieces as withheld holdouts. The partition, global-sign invariance, nested-prefix feasibility, and holdout-separation properties are exact finite facts. A replay on the locked $(N, H, z) = (512, 58, 5)$ spine, with $Q = 50, 60, 70, 90$, two kernel exponents, and three tolerances, gives 18 cases, 36 directional fits, and 54 normalizer rows. Budget and holdout preferences are concordant in 13 cases, discordant in 3, and unresolved in 2. All three discordances occur at $Q = 70 \rightarrow 90$ for exponent one, one at each tolerance. The result is a finite obstruction to promoting the earlier budget orientation to a stable extrapolation rule. The physical replay is explicitly numerical rather than a directed-rounding certificate, and no causal, asymptotic, arithmetic, or twin-prime conclusion is claimed.

1 Motivation and frozen object

TPC-305 introduced a fixed-operator target swap for overlapping moving prime shells, and TPC-306 arranged its four budgets as an operator-by-target table [1, 2]. That construction quantified interaction, but it did not put both operators in a single ambient row space. In particular, a transported target had a native completion outside the overlap that depended on the shell row. The next minimal question is therefore finite and geometric: can the common rows be held fixed while the exclusive rows are withheld from coefficient selection?

We retain the parent source and all numerical conventions. The source window is the integer interval $[257, 512]$, with height $H = 58$ and comparison cutoff $z = 5$. The shell spine is $Q = (50, 60, 70, 90)$; the kernel exponents are $e \in \{1, 2\}$; and the relative RMS tolerances are $\tau \in \{0.25, 0.5, 0.75\}$. The 17 profile cutoffs are

$$3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61.$$

No growing- N or growing- Q assertion is hidden in this notation.

For an adjacent pair write S_L, S_R for the locked prime shells and set

$$U = S_L \cup S_R, \quad O = S_L \cap S_R, \quad E_L = S_L \setminus O, \quad E_R = S_R \setminus O. \quad (1)$$

The physical matrix V_U has one row for every prime in U , generated by the same literal deleted-diagonal kernel formula. Thus the fit and both holdouts share one ambient row ordering. The source profile Gram matrix is denoted by M_k after restriction to the first k profile columns.

2 A directional holdout protocol

Let a_L and a_R be the source-first binary labels on S_L and S_R . The overlap alignment sign is

$$\sigma = \begin{cases} +1, & \sum_{p \in O} a_L(p)a_R(p) \geq 0, \\ -1, & \text{otherwise,} \end{cases} \quad (2)$$

and write $a_R^\sigma = \sigma a_R$. It is tempting to define one target on U by combining the two vectors coordinate by coordinate. That is not a consequence of the data: the two vectors can disagree on O , and an arbitrary choice would change the estimand. We consequently define two directional tasks instead:

$$b_L = a_L|_O, \quad h_L = a_L|_{E_L}, \quad b_R = a_R^\sigma|_O, \quad h_R = a_R^\sigma|_{E_R}. \quad (3)$$

For a direction $X \in \{L, R\}$ and a profile prefix of length k , let $V_{O,k}$ be the overlap rows of V_U and define the constrained source budget

$$B_{k,\tau}(b_X) = \min_c c^\top M_k c \quad \text{subject to} \quad \|V_{O,k}c - b_X\|_2 \leq \tau \|b_X\|_2. \quad (4)$$

The first feasible prefix for the two directions is k_L and k_R . We compare both at $k = \max(k_L, k_R)$, and predict on the complete union matrix. The exclusive holdout loss is

$$H_X = \frac{1}{|E_X|} \sum_{p \in E_X} ((V_{U,k}c_X)[p] - h_X(p))^2. \quad (5)$$

The reported ratios are right-over-left, both for the source budgets and for H_R/H_L . An interval wholly below 0.9 is called right-lower, one wholly above 1.1 left-lower, and the remaining interval unresolved.

Proposition 1 (Finite partition and holdout separation). *For finite S_L, S_R , (1) is a pairwise-disjoint partition of U . For each direction, the minimization (4) does not consult rows in E_L or E_R ; those rows enter only through (5).*

Proof. An element of $S_L \setminus O$ cannot lie in O , and similarly on the right. If an element lay in both exclusive differences, it would lie in both shells and hence in O , a contradiction. Every element of either shell is in the overlap or in exactly one exclusive difference, which proves the partition. The objective and constraint in (4) contain only $V_{O,k}$, b_X , M_k , and τ ; the exclusive rows occur only in the prediction and loss in (5). $\square$

Proposition 2 (Sign and common-prefix invariances). *Negating a directional overlap target and its matching holdout target leaves its budget and squared holdout loss unchanged. If the two first feasible prefixes are k_L, k_R , then $k = \max(k_L, k_R)$ is feasible for both.*

Proof. If c is feasible for b , then $-c$ is feasible for $-b$, with the same quadratic cost and residual norm. The prediction and holdout label are both negated, so every holdout residual is negated and its square is unchanged. For the prefix statement, pad a feasible coefficient vector at prefix j by zeros to any prefix $k \geq j$. The represented overlap vector is unchanged, so feasibility is nested; applying this to both directions proves the claim. $\square$

Remark 1 (What is not identified). *The directional construction shares V_U and the overlap fit geometry, but it does not manufacture a common target on U . Moreover, the labels are inherited from TPC-302's physical-Gram-dependent source-first construction. Thus the holdout is a diagnostic of finite completion stability, not a causal intervention separating target from operator.*

Table 1: Budget/holdout agreement by shell transition.

transition	cases	concordant	discordant	unresolved
50 $\rightarrow$ 60	6	5	0	1
60 $\rightarrow$ 70	6	5	0	1
70 $\rightarrow$ 90	6	3	3	0
total	18	13	3	2

3 Finite replay and obstruction

The producer reconstructs the locked source rows, builds the six union shells, solves (4) for all 18 (Q_L, Q_R, e, τ) cells, and records three positive normalizers: the source norm, the profile-trace mean, and the first-profile norm. The physical rows are assembled by a vectorized float64 evaluation of the literal formula and decimalized before the 70-digit frontier solve. A relative padding of 10^{-5} is used. This data path is intentionally called *numerically reproduced*, not formally certified by directed rounding.

The independent checker does not import the producer. It loads the frozen TPC-268 engine, rebuilds the source profiles and union rows with NumPy, solves the frontier independently, and checks the stored enclosures, prefixes, partitions, ratios, and classes. A rational stress suite separately tests the finite algebraic identities.

Table 1 is the central finite result. Budget preferences alone are strict in all 18 cells: right-lower in 13 and left-lower in 5. The exclusive holdout is right-lower in 13, left-lower in 3, and unresolved in 2. The three discordances are not diffuse: they are exactly the three exponent-1 cells on the final 70 $\rightarrow$ 90 transition, at $\tau = .25, .5, .75$. For example, the right/left budget ratios there are approximately 1.152, 1.266, and 0.888, while the corresponding holdout ratios are approximately 0.280, 0.511, and 1.156. Hence each pair crosses the strict preference threshold in the two diagnostics.

At the middle transition the exponent-1, $\tau = .75$ cell is unresolved on the holdout despite a strict budget ratio near 0.108; the other unresolved cell is at 50 $\rightarrow$ 60, exponent 1, $\tau = .75$. These cells are useful warnings against reading a budget frontier as an automatic prediction of exclusive completion error.

4 Interpretation and route status

The positive result is a clean finite protocol: a common ambient operator can be formed, overlap fitting can be separated mathematically from exclusive testing, and global sign and prefix choices can be audited without ambiguity. The strongest negative result is the localized discordance atlas. Even after sharing the ambient operator, three budget preferences reverse when judged on the withheld native exclusive pieces. This is a finite obstruction to a completion-invariant preference claim on the declared spine.

The result does not pay the arithmetic L^2 gate, a uniform growing-budget estimate, or any fixed-power credit. It does not prove full Route-B Gate B, and it says nothing by itself about the existence of infinitely many twin primes. The Session-named Route-A/Route-B evaluator files are absent from the checkout, so no official evaluator pass is asserted; the local fail-closed assessment is recorded in the accompanying route note.

The immediate next question is adversarial rather than promotional: replace the native exclusive completion by explicitly bounded alternative completion envelopes and perturb the profile prefix around the selected k . If the three discordances persist, they become a stronger scoped obstruction; if they disappear, the native completion is identified as the missing structure.

Reproducibility

The canonical result is `results/tpc307_certificate.json`. The source, independent replay, stress suite, theorem ledger, and Bridge-B checker are included in the project directory. All numerical statements above are finite observations under the locked protocol and should not be extrapolated to a moving asymptotic regime.

References

- [1] Liang Wang. A fixed-operator counterfactual for overlapping prime-shell labels, 2026. Local TPC-305 release in the prime-dynamics research repository.
- [2] Liang Wang. Two-way log-budget decomposition of prime-shell operator interaction, 2026. Local TPC-306 release in the prime-dynamics research repository.